

Tilak Kumar

Senior Software Engineer

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Experience

Avacend Inc	Remote
Senior Software Engineer - Contract	Dec 2024 - Present
- Built Python data pipelines with OAuth2-authenticated WorxHub API integration, automatic token refresh, and three-tier extraction strategy (cumulative, delta, on-demand) for incremental data synchronization - Designed ClickHouse analytics schema with ReplacingMergeTree engine, 100K-row batch inserts across 8 parallel threads with exponential backoff retry logic - Implemented text-to-SQL generation pipeline using GPT-4o converting natural language queries to ClickHouse SQL, with AI-powered result interpretation for human-readable work order descriptions - Built ML pipeline using SentenceTransformer embeddings, HDBSCAN clustering, and UMAP dimensionality reduction for work order pattern analysis with Pinecone vector database integration - Developed 4 production dashboards with 10+ KPIs including SLA compliance tracking, shift-based analytics with dynamic ClickHouse query generation, and DragonflyDB caching with extraction-date-aware invalidation - Architected multi-tenant FastAPI backend with three-tier RBAC, JWT authentication, and async ClickHouse operations using ThreadPoolExecutor to prevent event loop blocking	
Constient Global Solutions	Nov 2022 – Dec 2024
Senior Software Engineer	
- Founding engineer of the startup project; designed and implemented scalable REST API backend microservices using GoLang, optimizing for high performance and efficiency - Architected a distributed log management system using StarRocks DB and Kafka, improving troubleshooting efficiency by 40% - Implemented high-performance gRPC communication protocols between services using Python and GoLang - Created SDKs and integrated OpenAI APIs for advanced AI-driven backend features - Managed deployment infrastructure using Docker and Kubernetes for microservices architecture - Optimized database performance across PostgreSQL, ClickHouse, and StarRocks implementations - Developed and maintained robust CLI tools with GoLang for streamlining backend operations - Architected responsive front-end for the log management system using Next.js and TailwindCSS, with real-time data visualization and AI-recommended filters - Established frontend testing practices using Jest, achieving 90% test coverage	

Projects

Amour - Dating App (Full Stack)	2024 - Present
- Architected NestJS microservices with gRPC transport layer supporting 16 proto service definitions, Envoy proxy integration, and hexagonal architecture with interface-based dependency injection across 68+ services - Designed 68-model Prisma schema (1,725 lines) with 141 BTREE indexes and denormalized chat schema caching, achieving 50x faster chat list rendering by eliminating subqueries - Engineered dual authentication strategy — GatewayHelper (10-30x faster, reads pre-validated user_id from Envoy header) vs SuperTokens network verification for critical operations, with AES-256-GCM encryption for sensitive identity data - Built 4-tier priority queuing system (CRITICAL/HIGH/NORMAL/LOW) with presence-aware routing, circuit breaker pattern with DB failover, and burst detection for notification batching across 16 RabbitMQ topic exchanges - Eliminated N+1 queries in recommendation service — pre-fetches candidates, scores, and boost states in 6 parallel queries instead of 200+ sequential lookups, with multi-factor scoring (distance 0.30, compatibility 0.40, activity 0.10) - Built perceptual hash-based photo deduplication with O(k) prefix filtering reducing search space before Hamming distance calculation, integrated with AWS Rekognition for content moderation and minor detection - Architected horizontally-scalable WebSocket gateway using Socket.io with Redis pub/sub adapter for cross-pod message delivery, typing indicators, and offline message queuing via RabbitMQ - Orchestrated FluxCD GitOps pipeline with image automation, multi-stage Docker builds (ARM64/x86_64, 614MB production image), Kubernetes rolling updates with zero-downtime deployments, and Trivy vulnerability scanning in CI - Built React Native mobile app with Redux Toolkit, Redux Persist, and Redux Saga for complex state management and secure session persistence - Implemented gRPC-Web communication with TypeScript type safety, handling protobuf deserialization and nested error patterns - Integrated Agora RTM for real-time messaging with offline queuing and message synchronization	

Skills

Backend	Frontend	Infrastructure & AI/ML
Python, Node.js (TypeScript), GoLang, PostgreSQL, ClickHouse, StarRocks, MongoDB, Neo4j, Redis, DragonflyDB, Kafka, RabbitMQ, REST, gRPC, WebSocket	TypeScript, Jest, TailwindCSS, Redux Toolkit, React Native, Next.js, React.js, Figma	Docker, Kubernetes, GitHub Actions, FluxCD, Trivy, GCP, AWS, Cloudflare, SentenceTransformers, Pinecone

Education

Dayananda Sagar Institution	Diploma
Computer Science	